

week. Information on selection of crops and their varieties, proper sowing time, pest and disease control, choosing optimum time and dose of fertiliser, post-harvesting advice, irrigation management and other weather-based farm decision making is also a part of their service.

After launching its service in 21 districts of Punjab and 20 districts of Haryana, Handygo is now expanding its services all across India with Andhra Pradesh as its major focus, where it is covering 23 districts. The company will also expand its focus in Gujarat, West Bengal and Maharashtra.

Dye-sensitised thin-film solar cells

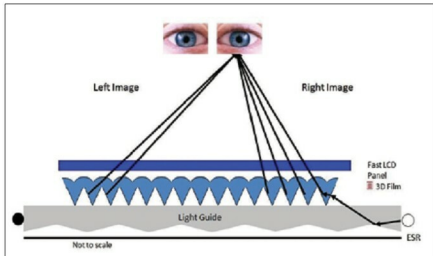
The first commercial product to incorporate dye-sensitised thin-film solar cells will soon be in the market. Back-packs coated with cheap, lightweight, and flexible solar cells for on-the-go recharging of portable gadgets, were recently unveiled at the Hong Kong Electronics Fair.

The solar cells made by UK-based G24 Innovations, are based on technology invented by a chemistry professor at the École Polytechnique Fédérale de Lausanne in Switzerland.

In this type of solar cell, dye-coated semiconductor nanocrystals are sandwiched between glass panels or embedded in plastic along with an electrolyte. The dye absorbs light and creates electrons which are transferred to the semiconductor and then out into a circuit. Dye-sensitised cells have lower light-to-electricity conversion efficiencies than the thin-film solar cells. They are considerably cheaper to manufacture and can also be printed on flexible surfaces.

Mobile phones to go 3D

A new thin-film technology developed by 3M would enable mobile devices such as cellphones to show 3D images without the need for special glasses. Vikuiti 3D technology works by giving slightly different images to the view-



3M Vikuiti 3D display film

er's left and right eyes. If the device is held still, the viewer can experience an 'auto-stereoscopic' effect.

In Vikuiti 3D, prism-shaped reflective structures are embedded on the back of a polymer film and tiny microlenses are patterned on the front. Together these components steer lights through a liquid crystal display (LCD) in front of the film. Light passes through the film from two light-emitting diodes (LED). The light from each LED bounces off a waveguide and strikes the film at a different angle, causing the embedded optics of the film to steer the light in two different directions. As each beam of light passes through an LCD display, it shows a slightly different image to each eye.

Home robots can be hacked

Several commercial home robots with small cameras, microphones and other audio-visual surveillance features have been found less secured.

Researchers at the University of Washington studied 2008 models of the Erector Spykee, WowWee's RoboSapien and Rovio robots and found security holes that include unencrypted audio-visual streams, unencrypted user names and passwords for accessing and controlling the bots and tricks for taking over the robots remotely.

According to the researchers, these

vulnerabilities mean that someone might be able to drive your robot around your home, look around the house and listen to conversations. The researchers point out that better security and privacy safety measures will need to be taken as home robots are becoming more common. To stay protected, they recommend keeping networks and robot control encrypted, avoiding remote access, and also turning off the robots when they're not in use.

Touchless 3D fingerprinting

Researchers at the University of Kentucky have created a new non-contact, 3D fingerprinting system which could make spotting criminals easily. By projecting patterns of light onto a finger and analysing the image, researchers were able to create a more accurate print than those made with ink or sensor plates.

Researchers say the system is more efficient than traditional fingerprinting and significantly reduces the number of incorrect matches.

The device works by projecting a series of striped lines onto a finger in a process called structured light illumination (SLI). A 1.4-megapixel camera automatically captures images of the lines as they wrap around the finger. By analysing the way each line rises and falls, the software builds a 3D model of the surface of the finger in less than a second.

Check efytimes.com for more news, daily